

Polynomial Regression: Model Results

Machine Learning Assignment 1 | Two Geothermal Problems

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1. Introduction and Data

This report uses polynomial regression on two datasets. **Problem 1 (var1)** predicts turbine power score from six input values using degrees up to 10. **Problem 2 (var2)** predicts thermal score from three location values using degrees up to 20. Each problem has 1,000 training rows and 1,000 test rows.

2. Method Used

I tested several polynomial models and compared their errors:

- **Polynomial features:** The model uses powers of each input and their interactions up to the selected degree.
- **Scaling:** The expanded features were scaled so that large powers did not dominate the model.
- **Models:** I compared Linear Regression, Ridge, Lasso, and Elastic Net with penalty values 0.01, 0.1, 1.0, and 10.0.
- **Validation:** I used shuffled 5-fold cross-validation with seed 42. Models were compared using average R2 and MSE.

3. Best Models

The best model for each problem was selected from all tested choices:

Problem	Phase	Inputs	Selected Algorithm	Opt. Degree	Penalty alpha	CV R2	CV MSE
var1	Steam Turbine Power	6	Lasso Regression	5	0.01	0.9703	0.3013
var2	Reservoir Thermal Map	3	Ridge Regression	12	1.0	0.9935	0.2566

4. Error by Polynomial Degree

The plots show how the validation error changes as the polynomial degree increases. Very small degrees are too simple, while very large degrees can fit noise. Ridge and Lasso help control this problem.

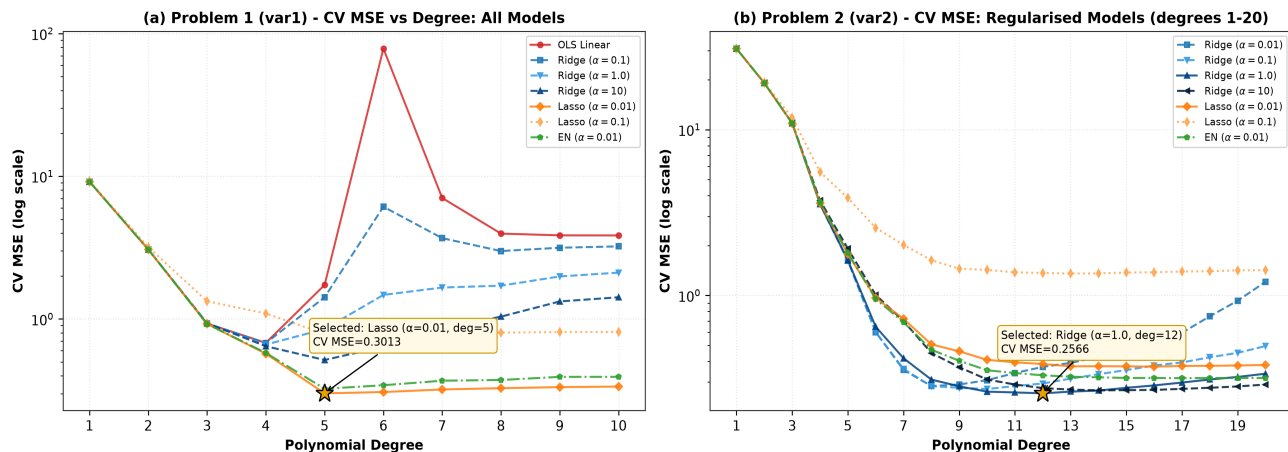


Figure 1: Average validation MSE for different degrees. Ridge gives the best result for Problem 2 at degree 12.

5. Choosing the Degree

For Problem 1, Lasso at degree 5 has the lowest validation MSE of 0.3013. After degree 5, the error does not improve enough to justify a more complicated model.

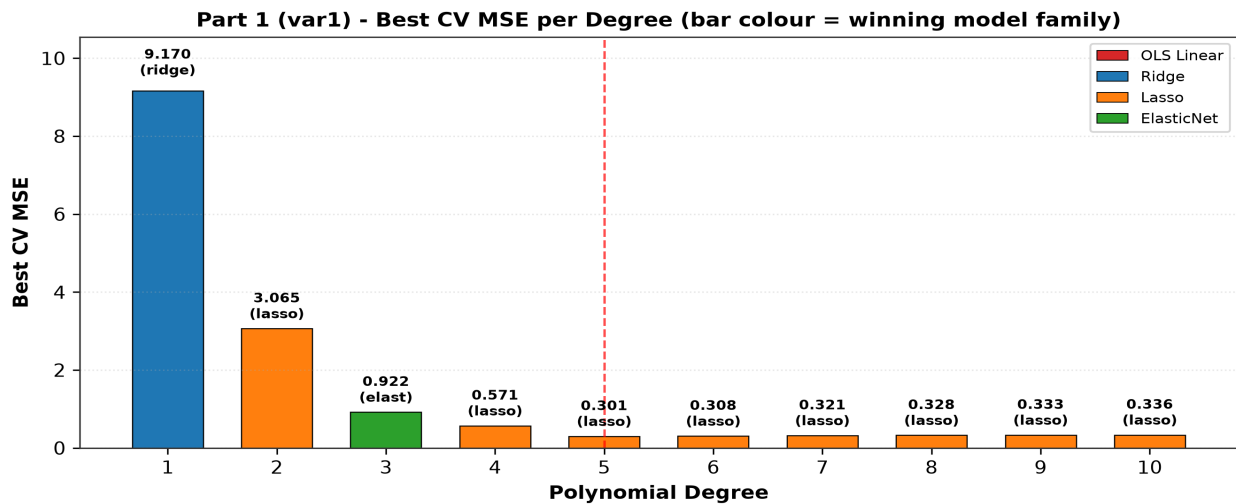


Figure 2: Lowest validation MSE found at each degree for Problem 1. The best result is Lasso at degree 5.

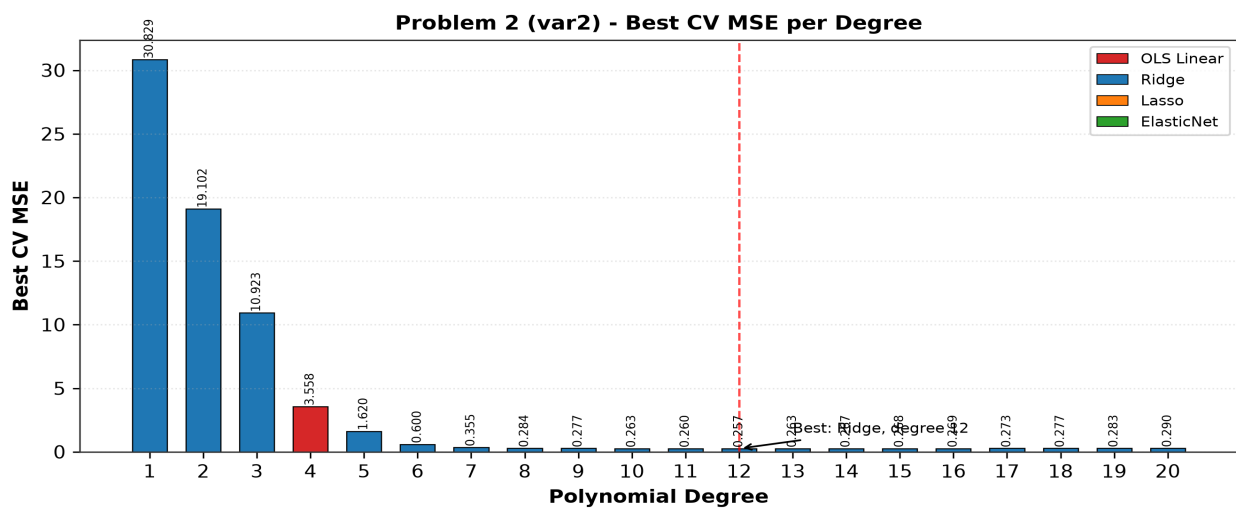


Figure 3: Lowest validation MSE found at each degree for Problem 2. The best result is Ridge at degree 12.

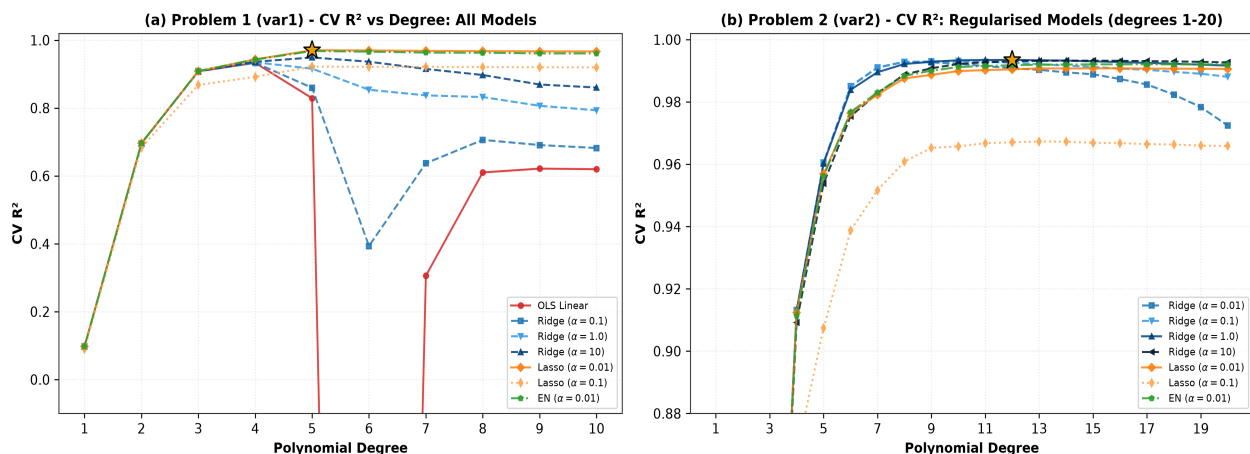


Figure 4: Average validation R² by degree for both problems and all tested model settings.

Important observation: Ordinary Linear Regression becomes unstable at high degrees. Its Problem 1 error rises from 0.6799 at degree 4 to 78.4 at degree 6. For Problem 2, its error reaches 1160.8 at degree 20. Ridge and Lasso give more stable results.

6. Model Predictions

The next plot compares predicted values with the real training values from cross-validation. Points close to the diagonal line show good predictions. The colours show the size of the prediction error.

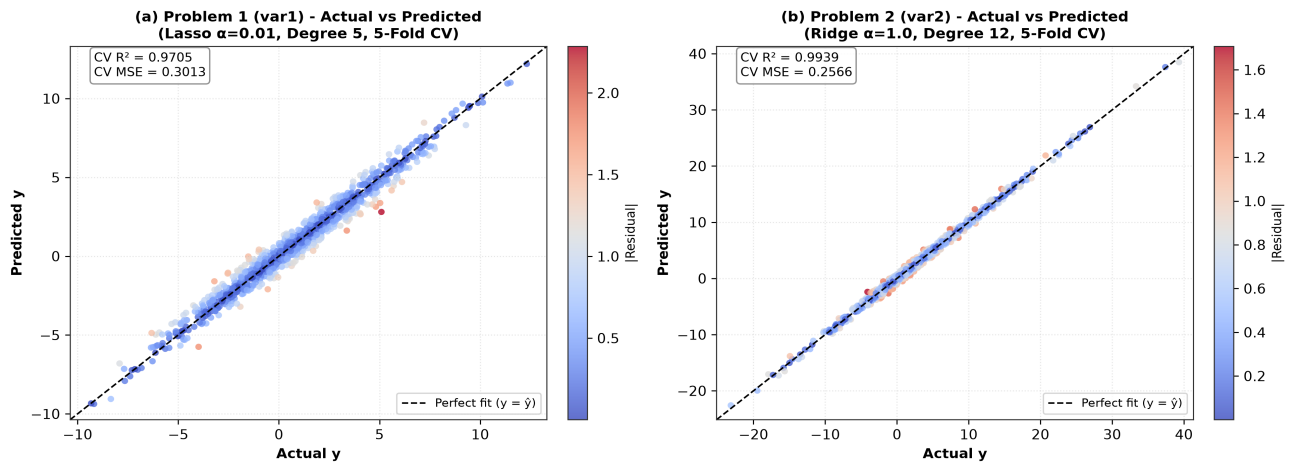


Figure 5: Predicted values compared with actual values. Problem 1 gives $R^2 = 0.9705$ and Problem 2 gives $R^2 = 0.9939$.

7. Prediction Errors

The residual plot checks whether the errors are spread evenly. For Problem 1, the errors are centred close to zero and have a standard deviation of about 0.549.

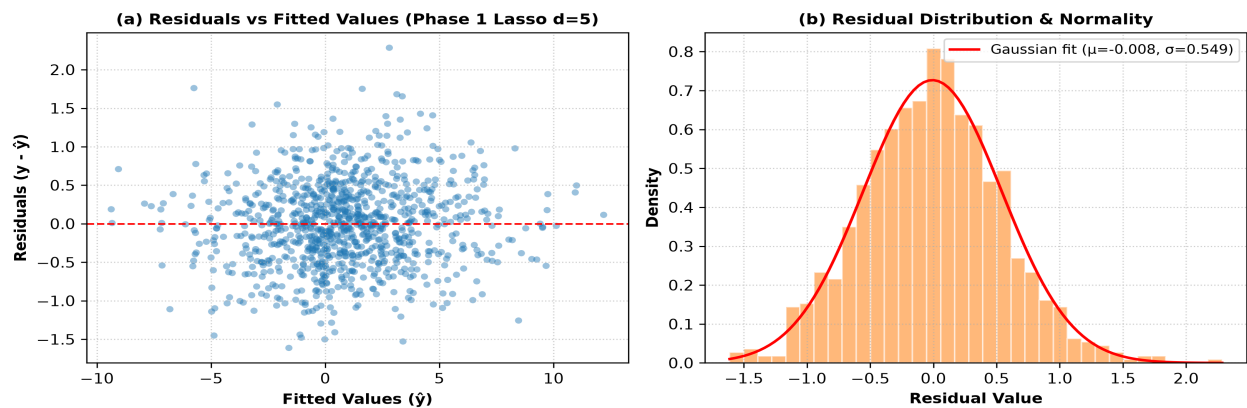


Figure 6: Prediction errors for Problem 1 using Lasso degree 5.

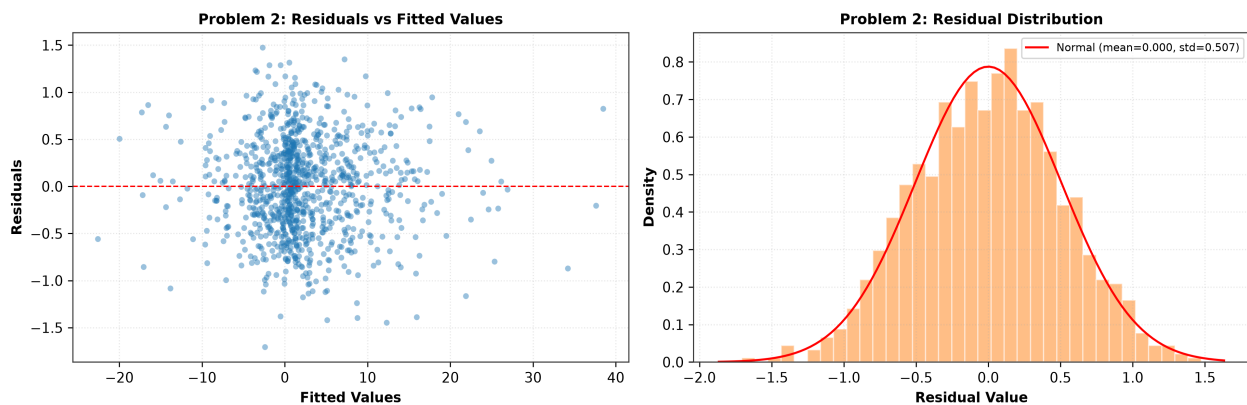


Figure 7: Prediction errors for Problem 2 using Ridge degree 12.

8. Interpretation of Results

The two problems have different input data, so different models work best:

- **Problem 1:** Lasso with degree 5 is best. It can ignore some less useful polynomial terms and gives a low error.
- **Problem 2:** Ridge with degree 12 is best. It keeps the useful terms but reduces the effect of very large coefficients.
- **Repeated points:** Some Problem 2 coordinates appear more than once. Their median variance is about 0.255, which is close to the final validation MSE of 0.2566.

9. Final Files

The final models were trained again using all 1,000 training rows. The prediction files contain one column named y and keep the same order as the test rows:

Phase	Dataset	Winning Configuration	Output File	Row Count	Target Col.
Phase 1	Turbine Calibration	Lasso (deg=5, alpha=0.01)	BT2024259_pred_var1.csv	1000	y
Phase 2	Thermal Reservoir	Ridge (deg=12, alpha=1.0)	BT2024259_pred_var2.csv	1000	y

GitHub Code Repository: https://github.com/Kandi-Rohan/ML_assignment